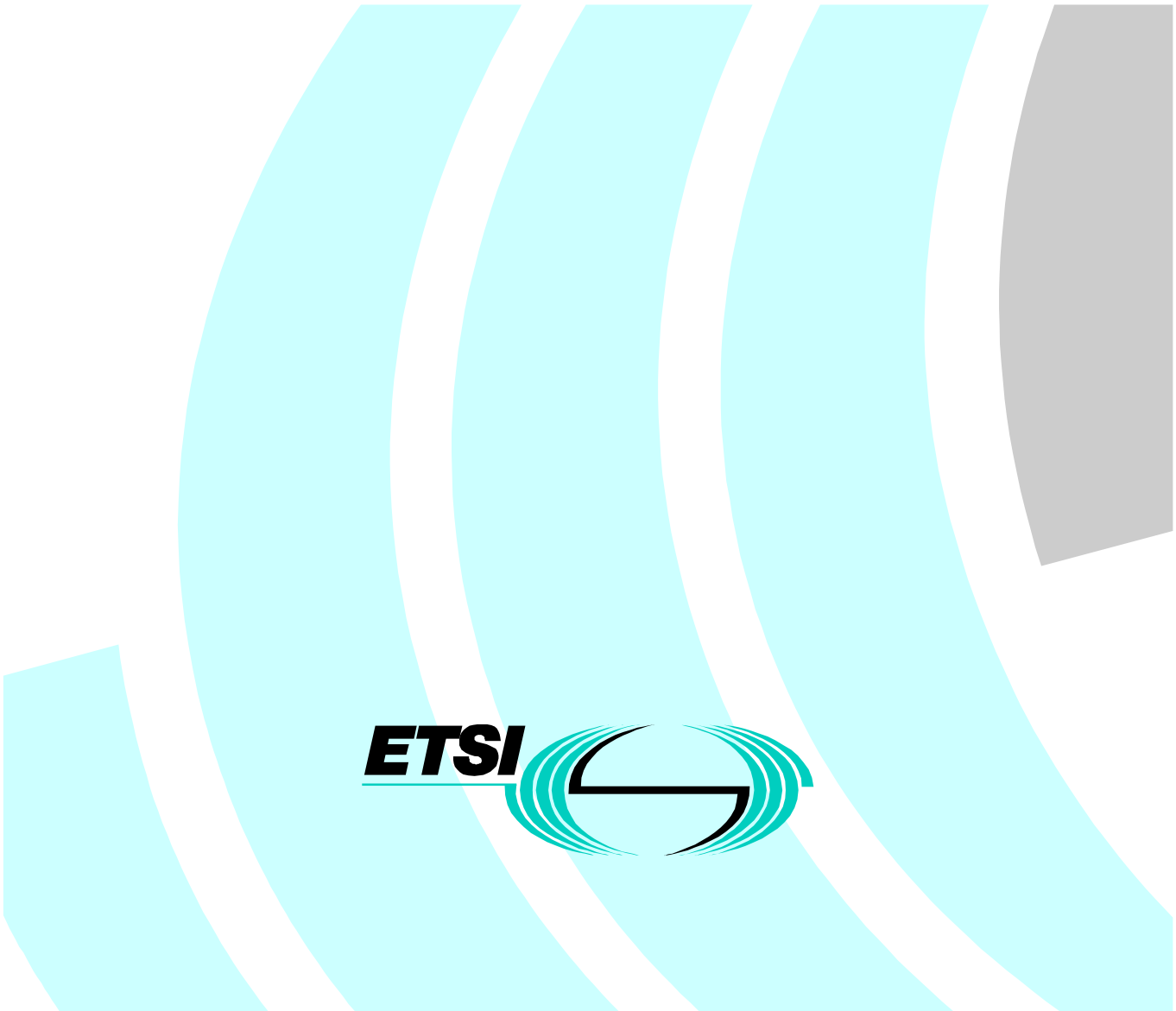
EN 300 188-3 V1.2.4 (1998-06)

European Standard (Telecommunications series)

Integrated Services Digital Network (ISDN); Three Party (3PTY) supplementary service; Digital Subscriber Signalling System No. one (DSS1) protocol; Part 3: Test Suite Structure and Test Purposes (TSS&TP) specification for the user



Reference
REN/SPS-05145-J2-3 (1qor0iqo.PDF)

Keywords
ISDN, DSS1, supplementary service, 3PTY,
testing, TSS&TP, user

ETSI

Postal address
F-06921 Sophia Antipolis Cedex - FRANCE

Office address
650 Route des Lucioles - Sophia Antipolis
Valbonne - FRANCE
Tel.: +33 4 92 94 42 00 Fax: +33 4 93 65 47 16
Siret N° 348 623 562 00017 - NAF 742 C
Association à but non lucratif enregistrée à la
Sous-Préfecture de Grasse (06) N° 7803/88

Internet
secretariat@etsi.fr
http://www.etsi.fr
http://www.etsi.org

Copyright Notification

No part may be reproduced except as authorized by written permission.
The copyright and the foregoing restriction extend to reproduction in all media.

© European Telecommunications Standards Institute 1998.
All rights reserved.

Contents

Intellectual Property Rights 4
Foreword 4
1 Scope 5
2 Normative references 5
3 Definitions 6
3.1 Definitions related to conformance testing 6
3.2 Definitions related to EN 300 188-1 6
4 Abbreviations 7
5 Test Suite Structure (TSS) 7
6 Test Purposes (TP) 8
6.1 Introduction 8
6.1.1 TP naming convention 8
6.1.2 Source of TP definition 8
6.1.3 TP structure 8
6.1.4 Test strategy 9
6.2 User TPs for 3PTY 9
6.2.1 Valid behaviour 9
6.2.2 Syntactically invalid behaviour 11
7 Compliance 12
8 Requirements for a comprehensive testing service 12
Annex A (informative): Changes with respect to the previous ETS 300 188-3 13
History 14

Intellectual Property Rights

IPRs essential or potentially essential to the present document may have been declared to ETSI. The information pertaining to these essential IPRs, if any, is publicly available for **ETSI members and non-members**, and can be found in SR 000 314: *"Intellectual Property Rights (IPRs); Essential, or potentially Essential, IPRs notified to ETSI in respect of ETSI standards"*, which is available **free of charge** from the ETSI Secretariat. Latest updates are available on the ETSI Web server (http://www.etsi.fr/ipr or http://www.etsi.org/ipr).

Pursuant to the ETSI IPR Policy, no investigation, including IPR searches, has been carried out by ETSI. No guarantee can be given as to the existence of other IPRs not referenced in SR 000 314 (or the updates on the ETSI Web server) which are, or may be, or may become, essential to the present document.

Foreword

This European Standard (Telecommunications series) has been produced by ETSI Technical Committee Signalling Protocols and Switching (SPS).

The present document is part 3 of a multi-part standard covering the Digital Subscriber Signalling System No. one (DSS1) protocol specification for the Integrated Services Digital Network (ISDN) Three Party (3PTY) supplementary service, as described below:

Part 1: "Protocol specification";

Part 2: "Protocol Implementation Conformance Statement (PICS) proforma specification";

Part 3: "Test Suite Structure and Test Purposes (TSS&TP) specification for the user";

Part 4: "Abstract Test Suite (ATS) and partial Protocol Implementation eXtra Information for Testing (PIXIT) proforma specification for the user";

Part 5: "Test Suite Structure and Test Purposes (TSS&TP) specification for the network";

Part 6: "Abstract Test Suite (ATS) and partial Protocol Implementation eXtra Information for Testing (PIXIT) proforma specification for the network".

The present version updates the references to the basic call specifications.

National transposition dates	
Date of adoption of this EN:	19 June 1998
Date of latest announcement of this EN (doa):	30 September 1998
Date of latest publication of new National Standard or endorsement of this EN (dop/e):	31 March 1999
Date of withdrawal of any conflicting National Standard (dow):	31 March 1999

1 Scope

This third part of EN 300 188 specifies the Test Suite Structure and Test Purposes (TSS&TP) for the User side of the T reference point or coincident S and T reference point (as defined in ITU-T Recommendation I.411 [7]) of implementations conforming to the stage three standard for Three Party (3PTY) supplementary service for the pan-European Integrated Services Digital Network (ISDN) by means of the Digital Subscriber Signalling System No. one (DSS1) protocol, EN 300 188-1 [1].

A further part of this EN specifies the Abstract Test Suite (ATS) and partial Protocol Implementation eXtra Information for Testing (PIXIT) proforma based on the present document. Other parts specify the TSS&TP and the ATS and partial PIXIT proforma for the Network side of the T reference point or coincident S and T reference point of implementations conforming to EN 300 188-1 [1].

2 Normative references

References may be made to:

a) specific versions of publications (identified by date of publication, edition number, version number, etc.), in which case, subsequent revisions to the referenced document do not apply; or

b) all versions up to and including the identified version (identified by "up to and including" before the version identity); or

c) all versions subsequent to and including the identified version (identified by "onwards" following the version identity); or

d) publications without mention of a specific version, in which case the latest version applies.

A non-specific reference to an ETS shall also be taken to refer to later versions published as an EN with the same number.

[1] EN 300 188-1 (V1.2): "Integrated Services Digital Network (ISDN); Three Party (3PTY) supplementary service; Digital Subscriber Signalling System No. one (DSS1) protocol; Part 1: Protocol specification".

[2] EN 300 188-2 (V1.2): "Integrated Services Digital Network (ISDN); Three Party (3PTY) supplementary service; Digital Subscriber Signalling System No. one (DSS1) protocol; Part 2: Protocol Implementation Conformance Statement (PICS) proforma specification".

[3] ISO/IEC 9646-1: "Information Technology - OSI Conformance Testing Methodology and Framework; Part 1: General Concepts".

[4] ISO/IEC 9646-2: "Information Technology - OSI Conformance Testing Methodology and Framework; Part 2: Abstract Test Suite specification".

[5] ISO/IEC 9646-3: "Information Technology - OSI Conformance Testing Methodology and Framework; Part 3: The Tree and Tabular Combined Notation".

[6] EN 300 196-1: "Integrated Services Digital Network (ISDN); Generic functional protocol for the support of supplementary services; Digital Subscriber Signalling System No. one (DSS1) protocol".

[7] ITU-T Recommendation I.411 (1993): "ISDN user-network interfaces - Reference configurations".

[8] EN 300 403-1: "Integrated Services Digital Network (ISDN); Digital Subscriber Signalling System No. one (DSS1) protocol; Signalling network layer for circuit-mode basic call control; Part 1: Protocol specification [ITU-T Recommendation Q.931 (1993), modified]".

[9] ITU-T Recommendation I.112: "Vocabulary and terms for ISDNs".

[10] CCITT Recommendation E.164: "Numbering plan for the ISDN era".

[11] ITU-T Recommendation I.210: "Principles of the telecommunication services supported by an ISDN and the means to describe them".

3 Definitions

For the purposes of the present document, the following definitions apply:

3.1 Definitions related to conformance testing

abstract test case: Refer to ISO/IEC 9646-1 [3].

Abstract Test Suite (ATS): Refer to ISO/IEC 9646-1 [3].

Implementation Under Test (IUT): Refer to ISO/IEC 9646-1 [3].

implicit send event: Refer to ISO/IEC 9646-3 [5].

lower tester: Refer to ISO/IEC 9646-1 [3].

point of control and observation: Refer to ISO/IEC 9646-1 [3].

Protocol Implementation Conformance Statement (PICS): Refer to ISO/IEC 9646-1 [3].

PICS proforma: Refer to ISO/IEC 9646-1 [3].

Protocol Implementation eXtra Information for Testing (PIXIT): Refer to ISO/IEC 9646-1 [3].

PIXIT proforma: Refer to ISO/IEC 9646-1 [3].

system under test: Refer to ISO/IEC 9646-1 [3].

Test Purpose (TP): Refer to ISO/IEC 9646-1 [3].

3.2 Definitions related to EN 300 188-1

3PTY active: An occurrence of the 3PTY supplementary service exists;

3PTY idle: An occurrence of the 3PTY supplementary service has not been requested;

Begin3PTY request: The 3PTY supplementary service has been requested by the served user;

Call Held auxiliary state: See EN 300 196-1 [6], subclause 7.1.2.

Call Reference (CR): See EN 300 403-1 [8], subclause 4.3.

component: See EN 300 196-1 [6], subclause 11.2.2.1.

End3PTY request: The served user wants to have a private conversation with one of the remote users.

Idle auxiliary state: See EN 300 196-1 [6], subclause 7.1.2.

Integrated Services Digital Network (ISDN): See ITU-T Recommendation I.112 [9], definition 308.

ISDN number: A number conforming to the numbering and structure specified in CCITT Recommendation E.164 [10].

invoke component: See EN 300 196-1 [6], subclause 11.2.2.1.

remote user: The two other users to the served user who are involved in the two calls that are joined together in a three-way conversation.

return error component: See EN 300 196-1 [6], subclause 11.2.2.1.

return result component: See EN 300 196-1 [6], subclause 11.2.2.1.

served user: The served user is the user who invokes the 3PTY supplementary service.

service; telecommunication service: See ITU-T Recommendation I.112 [9], definition 201.

supplementary service: See ITU-T Recommendation I.210 [11], subclause 2.4.

user: The DSS1 protocol entity at the User side of the user-network interface where a T reference point or coincident S and T reference point applies.

user (S/T): The DSS1 protocol entity at the User side of the user-network interface where a coincident S and T reference point applies.

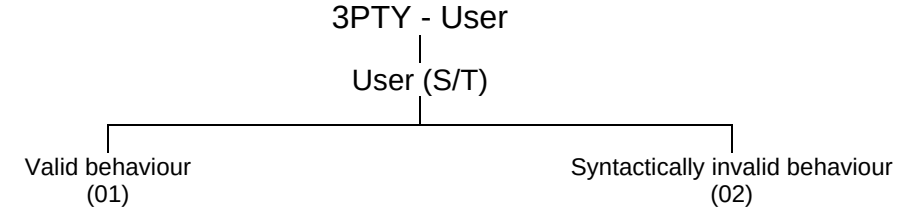
user (T): The DSS1 protocol entity at the User side of the user-network interface where a T reference point applies (User is the Private ISDN).

4 Abbreviations

For the purposes of the present document, the following abbreviations apply:

(Held)	Call Held auxiliary state
(Idle)	Idle auxiliary state
3PTY	Three Party
ATM	Abstract Test Method
ATS	Abstract Test Suite
CEI	Connection Endpoint Identifier
CR	Call Reference
CR1	CR for a call in the Active call state and Idle auxiliary state
CR2	CR for a call in the Active call state and Call Held auxiliary state
DSS1	Digital Subscriber Signalling System No. one
ISDN	Integrated Services Digital Network
IUT	Implementation Under Test
TP	Test Purpose
TSS	Test Suite Structure
U00	Idle call state
U10	Active call state
U11	Disconnect Request call state
U19	Release Request call state

5 Test Suite Structure (TSS)



NOTE: Numbers in brackets represent group numbers and are used in TP identifiers.

Figure 1: Test suite structure

6 Test Purposes (TP)

6.1 Introduction

For each test requirement a TP is defined.

6.1.1 TP naming convention

TPs are numbered, starting at 001, within each group. Groups are organized according to the TSS. Additional references are added to identify the actual test suite and whether it applies to the network or the user (see table 1).

Table 1: TP identifier naming convention scheme

Identifier: **<ss>_<iut><group>_<nnn>**				
<ss>	=	supplementary service:	e.g. "3PTY"	
<iut>	=	type of IUT:	U	User
			N	Network
<group>	=	group	2 digit field representing group reference according to TSS	
<nnn>	=	sequential number	(001-999)	

6.1.2 Source of TP definition

The TPs are based on EN 300 188-1 [1], clauses 9, 10 and 14.

6.1.3 TP structure

Each TP has been written in a manner which is consistent with all other TPs. The intention of this is to make the TPs more readable and checkable. A particular structure has been used and this is illustrated in table 2. This table should be read in conjunction with any TP, i.e. use a TP as an example to fully understand the table.

Table 2: Structure of a single TP

TP part	Text	Example
Header	<Identifier> *tab* <paragraph number in base ETS> *tab* <type of test> *tab* <condition> *CR.*	see table 1 subclause 0.0.0 valid, invalid, inopportune mandatory, optional, conditional
Stimulus	Ensure that the IUT in the <supplementary service state> and with CR1 in <basic call state> (<auxiliary state>) and with CR2 in <basic call state> (<auxiliary state>) <trigger> *see below for message structure* *or* <goal>	 3PTY Idle state and U10 (Idle), U10 (Held), etc. " receiving a XXXX message to request a ...
Reaction	<action> <conditions> *if the action is sending* *see below for message structure* <next action>, *etc.* and enters <supplementary service state> *and/or* and remains in the same call state(s) *or* and enters call state <state> with CR<number(s)>	sends, saves, does, etc. using en-bloc sending, ...
Message structure	<message type> message containing a *a)* <info element> information element with *b)* a <field name> encoded as *or* including <coding of the field> and *back to a or b,*	SETUP, FACILITY, CONNECT, ... Bearer capability, Facility, ...
NOTE: Text in italics will not appear in TPs and text between <> is filled in for each TP and may differ from one TP to the next.		

6.1.4 Test strategy

As the base standard EN 300 188-1 [1] contains no explicit requirements for testing, the TPs were generated as a result of an analysis of the base standard and the PICS specification EN 300 188-2 [2]. The criteria applied included the following:

- only the requirements from the point of view of the T or coincident Sand T reference point are considered;
- whether or not a test case can be built from the TP is not considered.

6.2 User TPs for 3PTY

NOTE: All calls have the same Connection Endpoint Identifier (CEI) value.

6.2.1 Valid behaviour

3PTY_U01_001 subclause 9.2.1.1, 2nd paragraph valid mandatory
Ensure that the IUT in the 3PTY Idle state, with CR1 in call state U10 (Idle) and CR2 in call state U10 (Held), to request the 3PTY supplementary service,
sends a FACILITY message containing a Facility information element with a Begin3PTY invoke component using the call reference of CR2 and enters the Begin3PTY Request state and remains in the same call states.

3PTY_U01_002 subclause 9.2.1.1, 5th paragraph valid mandatory
Ensure that the IUT in the Begin3PTY Request state, with CR1 in call state U10 (Idle) and CR2 in call state U10 (Held), on receipt of a FACILITY message for CR2 containing a Facility information element with a Begin3PTY return result component,
does not respond and enters the 3PTY Active state and remains in the same call states.

3PTY_U01_003 subclause 9.2.1.2, 3rd paragraph valid mandatory
Ensure that the IUT in the Begin3PTY Request state, with CR1 in call state U10 (Idle) and CR2 in call state U10 (Held), on receipt of a FACILITY message for CR2 containing a Facility information element with a Begin3PTY return error component,
takes no action and enters the 3PTY Idle state and remains in the same call states.

3PTY_U01_004 subclause 9.2.1.2, 5th paragraph valid mandatory
Ensure that the IUT in the Begin3PTY Request state, with CR1 in call state U10 (Idle) and CR2 in call state U10 (Held), on receipt of a FACILITY message for CR2 containing a Facility information element with a reject component,
takes no action and enters the 3PTY Idle state and remains in the same call states.

3PTY_U01_005 subclause 9.2.2.1 a) valid mandatory
Ensure that the IUT in the 3PTY Active state, with CR1 in call state U10 (Idle) and CR2 in call state U10 (Held), in order to clear the call associated with CR1,
sends a DISCONNECT message and enters the call state U11 for CR1 and remains in the same call state for CR2.

3PTY_U01_006 subclause 9.2.2.1 a) valid mandatory
Ensure that the IUT in the 3PTY Active state, with CR1 in call state U11 (Idle) and CR2 in call state U10 (Held), on receipt of a RELEASE message for CR1,
sends a RELEASE COMPLETE message and enters call state U00 for CR1, and sends a RETRIEVE message and enters the Retrieve Request auxiliary state for CR2 and remains in the same call state for CR2.

3PTY_U01_007 subclause 9.2.2.1 b) valid mandatory
Ensure that the IUT in the 3PTY Active state, with CR1 in call state U10 (Idle) and CR2 in call state U10 (Held), in order to clear the call associated with CR2,
sends a DISCONNECT message and enters the call state U11 for CR2 and remains in the same call state for CR1.

3PTY_U01_008 subclause 9.2.3.1, 1st paragraph valid mandatory
Ensure that the IUT in the 3PTY Active state, with CR1 in call state U10 (Idle) and CR2 in call state U10 (Held), in order to clear the calls associated with CR1 and CR2,
sends a DISCONNECT message and enters state U11 for CR1 and sends a DISCONNECT message and enters U11 for CR2.

3PTY_U01_009 subclause 9.2.3.1, 1st paragraph valid mandatory
Ensure that the IUT in the 3PTY Active state, with CR1 in call state U11 and CR2 in call state U11 (Held), on receipt of a RELEASE message for CR1,
sends a RELEASE COMPLETE message and enters state U00 for CR1 and does NOT send a RETRIEVE message for CR2 and remains in the same call state for CR2.

3PTY_U01_010 subclause 9.2.4.1 a) valid mandatory
Ensure that the IUT in the 3PTY Active state, with CR1 in call state U10 (Idle) and CR2 in call state U10 (Held), to establish a private communication with the remote user associated with CR2,
sends a FACILITY message for CR2 containing a Facility information element with an End3PTY invoke component and enters the End3PTY Request state and remains in the same call states.

3PTY_U01_011 subclause 9.2.4.1 a) valid mandatory
Ensure that the IUT in the End3PTY Request state (invoked for CR2), with CR1 in call state U10 (Idle) and CR2 in call state U10 (Held), on receipt of a FACILITY message for CR2 containing a Facility information element with an End3PTY return result component,
sends a HOLD message for CR1 and a RETRIEVE message for CR2 and enters the 3PTY Idle state and on completion of the Hold and Retrieve functions enters state U10 (Held) for CR1 and state U10 (Idle) for CR2.

3PTY_U01_012 subclause 9.2.4.1 b) valid mandatory
Ensure that the IUT in the 3PTY Active state, with CR1 in call state U10 (Idle) and CR2 in call state U10 (Held), to establish a private communication with the remote user associated with CR1,
sends a FACILITY message for CR1 containing a Facility information element with an End3PTY invoke component to the network and enters the End3PTY Request state and remains in the same call states.

3PTY_U01_013 subclause 9.2.4.1 b) valid mandatory
Ensure that the IUT in the End3PTY Request state (invoked for CR1), with CR1 in call state U10 (Idle) and CR2 in call state U10 (Held), on receipt of a FACILITY message for CR1 containing a Facility information element with an End3PTY return result component,
takes no action and enters the 3PTY Idle state and remains in the same call states.

3PTY_U01_014 subclause 9.2.4.2, 5th paragraph valid mandatory
Ensure that the IUT in the End3PTY Request state (invoked for CR1), with CR1 in call state U10 (Idle) and CR2 in call state U10 (Held), on receipt of a FACILITY message for CR1 containing a Facility information element with an End3PTY return error component,
takes no action and enters the 3PTY Active state and remains in the same call states.

3PTY_U01_015 subclause 9.2.4.2, 5th paragraph valid mandatory
Ensure that the IUT in the End3PTY Request state (invoked for CR2), with CR1 in call state U10 (Idle) and CR2 in call state U10 (Held), on receipt of a FACILITY message for CR2 containing a Facility information element with an End3PTY return error component,
takes no action and enters the 3PTY Active state and remains in the same call states.

3PTY_U01_016 subclause 9.2.4.2, 6th paragraph valid mandatory
Ensure that the IUT in the End3PTY Request state (invoked for CR2), with CR1 in call state U10 (Idle) and CR2 in call state U10 (Held) (where a three-way conversation exists), on receipt of a FACILITY message for CR2 containing a Facility information element with a reject component,
takes no action and enters the 3PTY Active state and remains in the same call states.

3PTY_U01_017 subclause 9.2.4.2, 6th paragraph valid mandatory
Ensure that the IUT in the End3PTY Request state (invoked for CR1), with CR1 in call state U10 (Idle) and CR2 in call state U10 (Held) (where a three-way conversation exists), on receipt of a FACILITY message for CR1 containing a Facility information element with a reject component,
takes no action and enters the 3PTY Active state and remains in the same call states.

3PTY_U01_018 subclause 9.2.5.1 a) valid mandatory
Ensure that the IUT in the 3PTY Active state, with CR1 in call state U10 (Idle) and CR2 in call state U10 (Held) on receipt of a DISCONNECT message for CR1,
sends a RELEASE message and enters call state U19 for CR1 and sends a RETRIEVE message and enters the Retrieve Request auxiliary state for CR2 and remains in the same call state for CR2.

3PTY_U01_019 subclause 9.2.5.1 b) valid mandatory
Ensure that the IUT in the 3PTY Active state, with CR1 in call state U10 (Idle) and CR2 in call state U10 (Held) on receipt of a DISCONNECT message for CR2,
responds with a RELEASE message and enters call state U19 for CR2 and remains in the same call state for CR1.

6.2.2 Syntactically invalid behaviour

3PTY_U02_001 subclauses 7.1 and 9.2.1.2 invalid mandatory
Ensure that the IUT in the Begin3PTY Request state, with CR1 in call state U10 (Idle) and CR2 in call state U10 (Held), on receipt of a FACILITY message for CR2 containing a Facility information element with an invalid Begin3PTY return result component,
sends a FACILITY message for CR2 containing a Facility information element with a reject component and remains in the same call states.

3PTY_U02_002 subclauses 7.1 and 9.2.1.2 invalid mandatory
Ensure that the IUT in the Begin3PTY Request state, with CR1 in call state U10 (Idle) and CR2 in call state U10 (Held), on receipt of a FACILITY message for CR2 containing a Facility information element with an invalid Begin3PTY return error component,
sends a FACILITY message for CR2 containing a Facility information element with a reject component and remains in the same call states.

3PTY_U02_003 subclauses 7.1 and 9.2.4.2 invalid mandatory
Ensure that the IUT in the End3PTY Request state (invoked for CR2), with CR1 in call state U10 (Idle) and CR2 in call state U10 (Held), on receipt of a FACILITY message for CR2 containing a Facility information element with an invalid End3PTY return result component,
sends a FACILITY message for CR2 containing a Facility information element with a reject component and remains in the same call states.

3PTY_U02_004 subclauses 7.1 and 9.2.4.2 invalid mandatory
Ensure that the IUT in the End3PTY Request state (invoked for CR2), with CR1 in call state U10 (Idle) and CR2 in call state U10 (Held), on receipt of a FACILITY message for CR2 containing a Facility information element with an invalid End3PTY return error component,
sends a FACILITY message for CR2 containing a Facility information element with a reject component and remains in the same call states.

3PTY_U02_005 subclauses 7.1 and 9.2.4.2 invalid mandatory
Ensure that the IUT in the End3PTY Request state (invoked for CR1), with CR1 in call state U10 (Idle) and CR2 in call state U10 (Held), on receipt of a FACILITY message for CR1 containing a Facility information element with an invalid End3PTY return result component,
sends a FACILITY message for CR1 containing a Facility information element with a reject component and remains in the same call states.

3PTY_U02_006 subclauses 7.1 and 9.2.4.2 invalid mandatory
Ensure that the IUT in the End3PTY Request state (invoked for CR1), with CR1 in call state U10 (Idle) and CR2 in call state U10 (Held), on receipt of a FACILITY message for CR1 containing a Facility information element with an invalid End3PTY return error component,
sends a FACILITY message for CR1 containing a Facility information element with a reject component and remains in the same call states.

7 Compliance

An ATS which complies with this TSS&TP specification shall:

a) consist of a set of test cases corresponding to the set or to a subset of the TPs specified in clause 6;

b) use a TSS which is an appropriate subset of the whole of the TSS specified in clause 5;

c) use the same naming conventions for the test groups and test cases;

d) maintain the relationship specified in clause 6 between the test groups and TPs and the entries in the PICS proforma to be used for test case deselection;

e) comply with ISO/IEC 9646-2 [4].

In the case of a) or b) above, a subset shall be used only where a particular Abstract Test Method (ATM) makes some TPs untestable. All testable TPs from clause 6 shall be included in a compliant ATS.

8 Requirements for a comprehensive testing service

As a minimum the Remote test method, as specified in ISO/IEC 9646-2 [4], shall be used by any organization claiming to provide a comprehensive testing service for user equipment claiming conformance to EN 300 188-1 [1].

Annex A (informative): Changes with respect to the previous ETS 300 188-3

The following changes have been done:

- conversion to EN layout;
- substitution of non-specific references to basic standards where the intention is to refer to the latest version.

History

Document history		
Edition 1	October 1996	Publication as ETS 300 188-3
V1.2.3	February 1998	One-step Approval Procedure OAP 9824: 1998-02-13 to 1998-06-12
V1.2.4	June 1998	Publication

ISBN 2-7437-2311-4
Dépôt légal : Juin 1998